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Supplementary information

Electrically driven random lasing from a modified Fabry–Pérot laser diode

Electrically driven random lasing from a modified Fabry-Perot laser diode

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Supplementary Material

Section 1. Non distributed feedback architectures

Lasing action of the modified laser diode presented in the main manuscript is based on non-distributed feedback random lasing. We resume here our previous results with similar architectures, describing them in the general context of laser science and proposing new possible implementations.

In Figure S1, we show our schematic vision of random lasers (RLs) with non-distributed feedback by comparing it with conventional lasing devices. Green, yellow, red and blue color code corresponds to pump, gain region, light emission and feedback elements (mirrors or scatters), respectively. We represent schematically four general classes of lasers, divided into devices with non-distributed or distributed feedback (columns) and ordered and disordered feedback elements (rows). We refer to scatters as disordered feedback elements, while mirrors and a periodic modulation of refractive index are considered as ordered feedback elements.

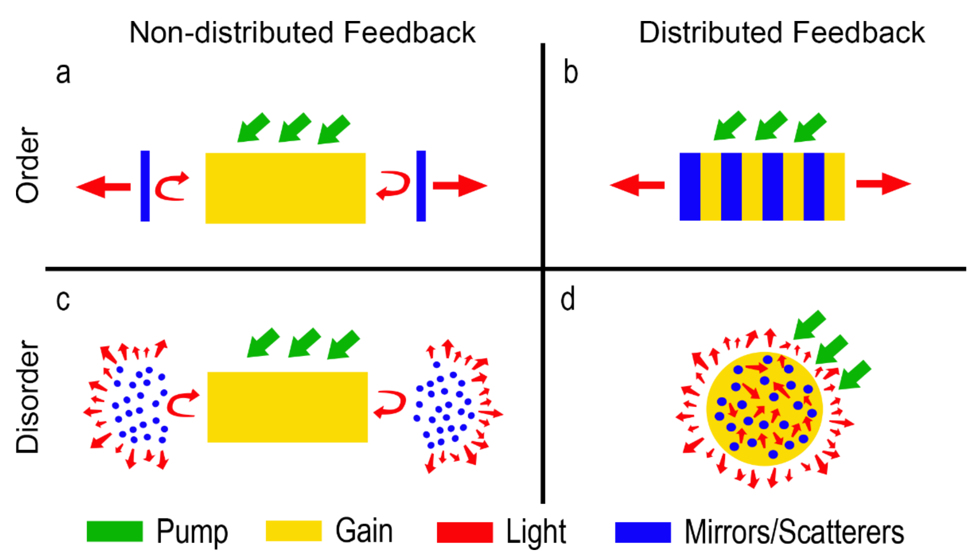

Figure S1. | Feedback and disorder in lasers. Fabry-Perot laser (a), DFB laser (b), RL with non-distributed (c) and with distributed feedback (d).

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In this view, Fabry-Perot (FP) lasers belong to the class of non-distributed feedback devices based on ordered elements (mirrors), see Fig. S1 (a), and distributed feedback (DFB) lasers belong to the class of distributed feedback devices based on ordered elements (periodic modulation of refractive index), see Fig. S1 (b). RLs with non-distributed feedback are obtained by simply substituting the ordered elements (mirrors) of a FP cavity with two disordered media (scatters), see Fig. S1 (c). We obtained the first demonstration of this kind of RLs with a device consisting of a liquid dye placed between two scattering volumes, i.e. agglomerations of titanium dioxide nanoparticles [1].

RLs with distributed feedback are the most diffused architecture of random lasing and consist of randomly distributed scatters inside the active medium of the device, see Fig. S1 (d). We consider that placing RLs with distributed and non-distributed feedback into this general view is a simple but relevant contribution, since this allows to draw analogies with conventional lasers, such as, for example, the introduction of a modified round trip condition, with frequency dependent amplitude and phase response of mirrors, for understanding why frequency modes are found at random spectral positions.

In Fig. S2, we show different alternative implementations of non-distributed feedback architectures with both ordered (mirrors) and disordered (scattering volumes or surfaces) elements. We previously demonstrated that not only scattering volumes, as in [1], but also scattering surfaces provide feedback for random lasing, see Fig. S2 (a). This was observed in devices where scattering surface were obtained by manual cut [2], pulsed laser ablation [3] or use of natural materials [4]. In Fig. S2(b), we show the architecture used in the present manuscript, demonstrating that both ordered and disordered feedback elements can be used in the same device with non-distributed feedback, i.e. in a laser diode where a scattering surface and a mirror provide feedback for lasing. In Fig. S2(c), we propose a structure in which a scattering volume and a mirror are used for feedback and lasing. In Fig. S2 (d) a scattering volume and a scattering surface are used. The structures shown in Fig. S2 are examples of alternatives structures that can be obtained by mixing ordered and disordered feedback elements. Additional architectures could also be obtained by using in the same device distributed/non-distributed feedback and ordered/disordered structures.

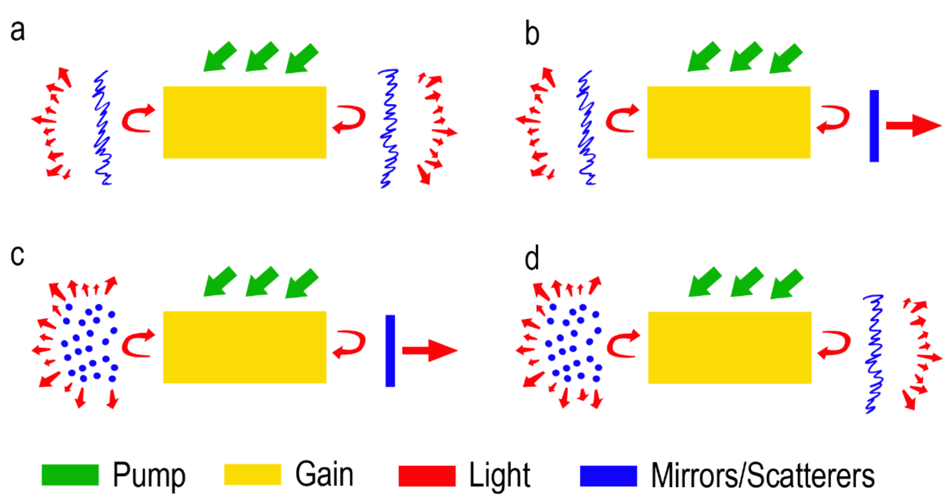

Figure S2. | Non-distributed feedback random lasers. RL with scattering surfaces (a) as in [2–4], RL with a scattering surface and a mirror (b) as in the present manuscript, RL with a scattering volume and a mirror (c) and RL with a scattering volume and a scattering surface (d).

Section 2. Original laser diode characterization

The emission of the original laser diode is characterized in terms of output power, spectral and spatial emission. A calibrated photodiode and a spectrometer are used for power and spectral measurements. Spatial coherence is measured with a double slit set-up, in which slits are 50 µm wide and placed at 500 µm of separation and interference fringes are imaged onto a CCD camera with a 20 cm cylindrical lens. Results are shown in Fig. S3. Lasing threshold (I_{TH} = 21 mA) and efficiency slope (η = 0.27 W/A) are obtained from output power versus bias current, see Fig. S3(a), blue dots and left axis. Spectral narrowing at threshold crossing is observed as the spectral full width at half maximum (FWHM) of the emission spectrum decrease from 16 nm to 0.3 nm, see Fig. S3(a), red dots and right axis. Typical emission spectra below and above threshold are plotted in Fig. S3(b), for 20.7 mA (black line) and 30 mA (red line), respectively. Equally spaced modes with free spectral range of 0.16 nm are found below threshold and a single main peak, with weak sidebands, is detected above threshold.

Interference fringes are obtained with the double slit set-up after summing up the detected intensities in CCD pixel columns parallel to the slits. Results for bias currents of 5 mA (black line) and 35 mA (red line) are shown in Fig. S3(c). The pronounced contrast between maxima and minima of curves in Fig. S3(c), corresponding to high values of spatial coherence, is quantified through the visibility γ, defined as $\gamma = (I_{MAX} - I_{MIN})/(I_{MAX} + I_{MIN})$ where I_{MAX} and I_{MIN} are the peak and trough intensities of the interference fringes, at the central peak, respectively. In Fig. S3(d), the visibility γ is plotted as a function of bias current and shows average values of 0.62 and 0.85, below and above threshold, respectively.

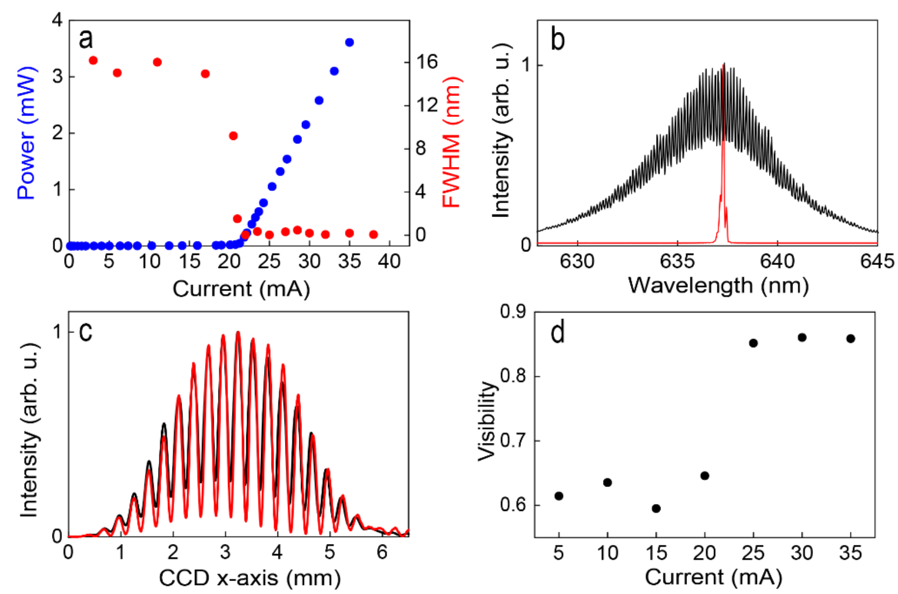

Figure S3. | Characterization of the original FP laser diode. (a) Power output (blue dots, left axis) and spectral at full-width at half-maximum (FWHM, red dots, right axis) versus injected current; (b) normalized emission spectrum below threshold, at 20.7 mA (black line) and above threshold, at 30 mA (red line). (c) Interference fringes measured with the two slits set-up, for currents of 5 mA (black curve) and 35 mA (red curve). (d) Fringe visibility γ, as defined in the main text, as a function of injected current.

The original laser diode is mounted in a TO-CAN package, which is removed prior to emission characterization and visual inspection presented in Fig. S4. In Fig. S4(a), the laser diode is found at the center of the image, mounted on a silicon substrate which is placed on the cathode terminal and heat sink. In Fig. S4(b), the laser dice (with the anode contact on top) and the silicon substrate are shown. The black area at the bottom in Fig. S4(c) and (d) is the silicon substrate and the thin active region can be seen close to the edge of the dice as a dark gray stripe when no current is applied, Fig. S4(c), and where a bright red emission is observed when applying 3 mA, Fig. S4(c).

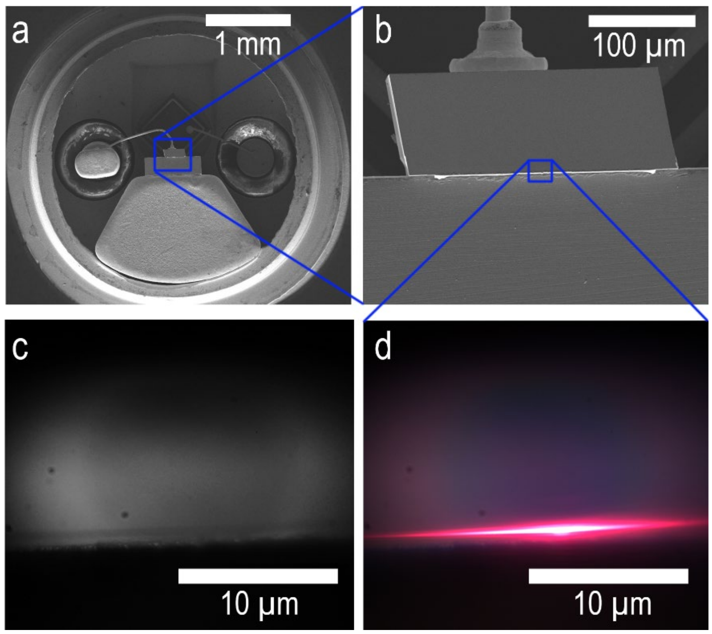

Figure S4. | The original laser diode. (a) SEM image of the opened can, the laser dice is in the center, placed on the silicon substrate and the heatsink. (b) SEM image of the laser dice, the anode contact on the top and the silicon substrate on the bottom of the image. Optical microscope image of the dice, when 0 mA (c) and 3 mA (d) are applied. The active region is the dark gray stripe, near the silicon substrate (black region at the bottom of the image).

Section 3. Surface analysis of the ablated emitting area

We provide an estimation of the disorder and surface roughness obtained after pulsed laser ablation of the front FP diode mirror by analyzing the images obtained by scanning electron microscopy (SEM).

The 256 gray levels of the SEM image of the emitting area are shown as in the original SEM image, Fig. S5(a), and as a 3D surface, Fig. S5(b). Horizontal (red line) and vertical scan directions cuts (black line) are considered in our analysis and plotted in Fig. S5 (c) and (d), respectively. Since we are interested in the roughness lateral autocorrelation, the magnitude (gray scale) representing secondary electrons

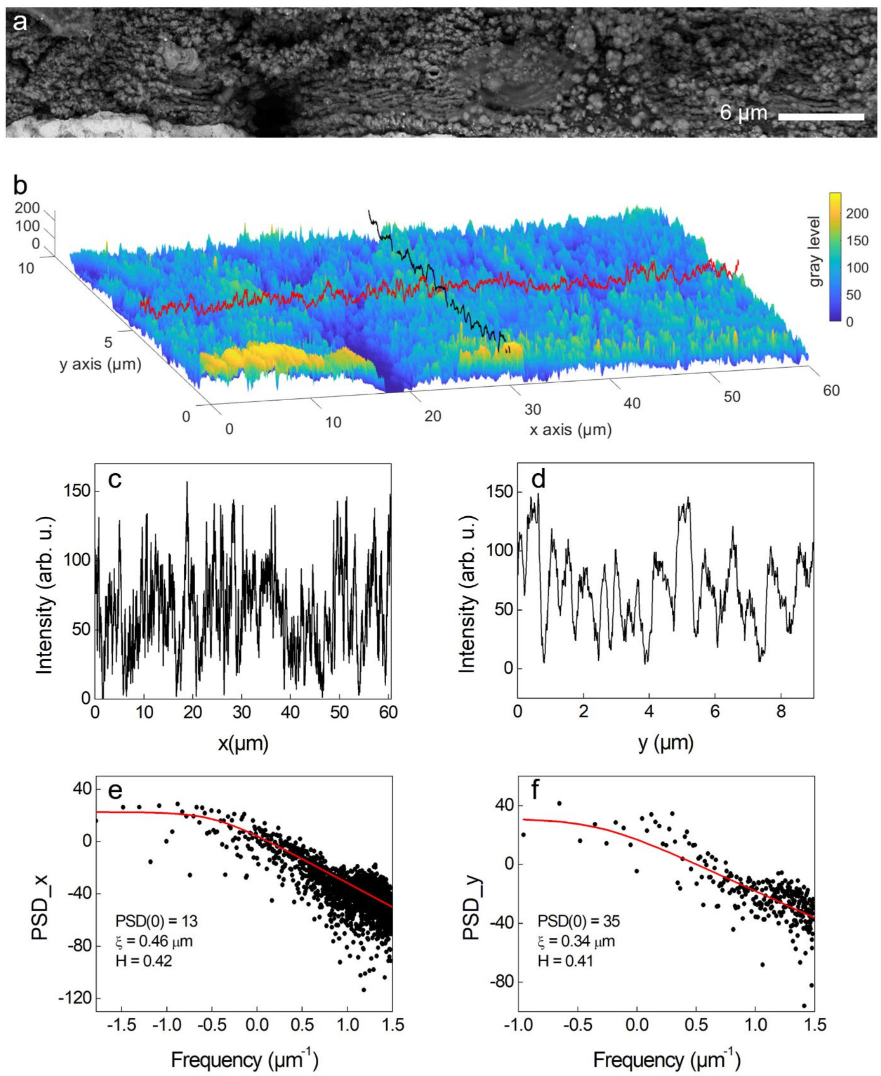

Figure S5. | Surface characterization of the modified emitting area. 256 gray level image of the emitting area of the modified laser diode obtained by SEM: original image (a) and 3D surface representation (b). Horizontal and vertical cross sections are highlighted by red and black lines in (b) and plotted in panels (c) and (d), respectively, the corresponding power spectral densities (PSD) are shown in (e) and (f).

intensity in the SEM image do not correspond to actual topographic heights, however it allows us to retrieve the in-plane autocorrelation length of the disordered surface scanned.

The vertical axis in Fig. S5 (c) and (d) corresponds to gray levels (from 0 to 256) of the SEM image along the red and black cuts, which provide information on lateral dimensions of the roughness profile induced by the ablation process. Knowledge of the absolute depth induced by the ablation process is only partial; however since very similar materials are involved in the active region (similar III-V semiconductors) variations of secondary electron emission can be expected to be fairly faithful to the shape of the surface. Plotted profiles show that local maxima and minima are spaced at sub-micron distances and profile fluctuations are random and unpredictable.

We calculated the power spectral density (PSD) [5] of the horizontal and vertical cuts by Fast Fourier Transform (FFT) and fitted it to the equation [6]:

$$PSD(f) = \frac{PSD(0)}{1+|2\pi f\xi|^{2H+1}}$$

where f is the spatial frequency, PSD(0) is the PSD at low frequencies, ξ is the correlation length in micrometers and H is the Hurst exponent.

Results of the fitting procedure are shown in Fig. S5 (e) and (f), for horizontal and vertical cuts, respectively. From the fitting procedure, the correlation length ξ is found to be 0.46 µm and 0.34 µm in the horizontal and vertical cross sections, respectively. This value corresponds to the distance at which two heights on a surface are no longer correlated (the autocorrelation function drops below 1/e) giving an approximate estimation of the average bump size. Although a precise characterization of the obtained rough surface is out of the scope of our contribution, with this simple analysis we aim to approximatively quantify the spatial lateral size of the roughness induced on the surface, which we estimate in the submicron range in the order of the wavelength of light emission.

Section 4. Far-field emission of the modified laser diode

In Fig. S6, the far-field emission [7] of the modified laser diode biased at 50 mA is shown in the wavevector space. In order to acquire this information an optical fiber with 100 µm diameter core is scanned across the back focal plane of the collecting lens so that each (X,Y) point corresponds to unique (ϑ,ϕ) angles of emission with respect to the original emission axis. A spectrum is acquired at each $I(x,y,\lambda) \rightarrow I(\vartheta,\phi,\lambda)$ position so that full hyperspectral characterization is obtained.

Figure S6(a) shows three points labelled as A, B and C, in Fourier space corresponding to different angles of emission. In Fig. S6 (b), spectra collected at these positions are plotted. Their lineshapes consist of randomly distributed peaks with sub-nanometer linewidths superimposed on a broad Gaussian background. At each point, the number and wavelength positions of the peaks are different, showing that each angle collects emission from different modes with different intensities thus giving rise to a specific spectral signature.

In order to illustrate the mapping of modes (i.e. in which directions is light of a given wavelength emitted) we select four narrow (0.25 nm wide) spectral regions centered at λ_1 = 629.3 nm, λ_2 = 630.6 nm, λ_3 = 632.4 nm and λ_4 = 633.2 nm (highlighted with green areas in the spectrum of Fig. S6 (b)) and extract from the $I(x,y,\lambda)$ parametric matrix four frequency resolved intensity maps $I(\vartheta,\phi\ \lambda_i)$, i = 1, …, 4 of the far-field distribution, by integrating spectra in those 0.25 nm wavelength windows around the concerned λ_i. Prior to evaluating the spectral integration, we subtracted a Gaussian background to each spectrum in order to remove the contribution of the amplified spontaneous emission that do not couple with resonant narrow modes. The Gaussian contribution was obtained from local minima of the measured spectrum

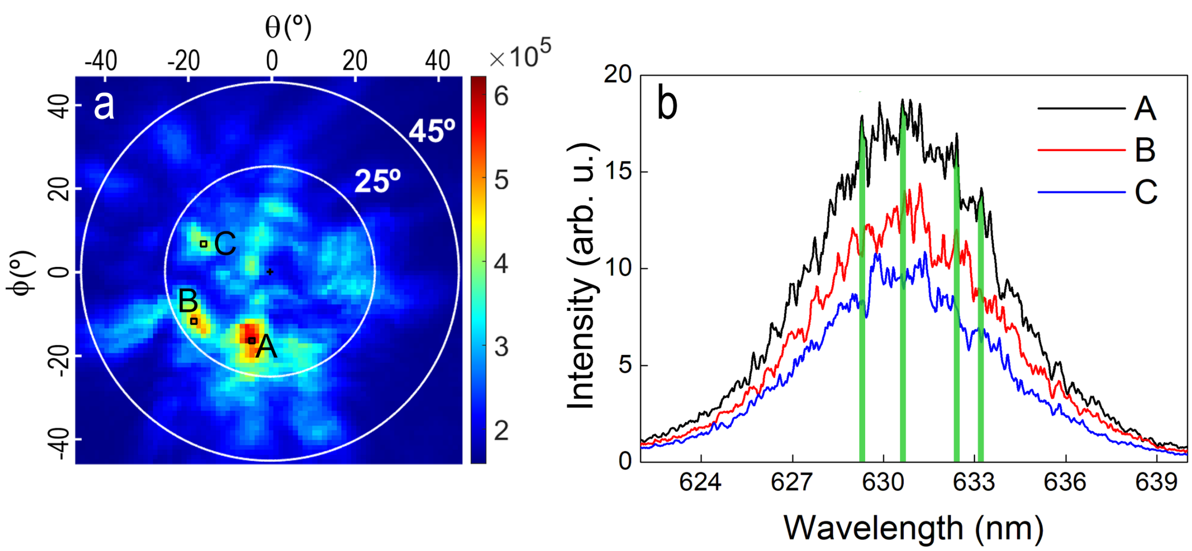

Figure S6. | Spectrally resolved far-field emission of the modified laser diode. (a) Intensity distribution on the far-field plane with injection current of 50 mA, obtained by integrating the spectrum collected at each point of the (ϑ,ϕ) plane over the entire wavelength detection range; the three black squares labelled A, B and C indicate the positions where spectra shown in panel (b) are collected. (b) Spectra collected at positions A (black curve), B (red curve) and C (blue curve): the green areas indicate the wavelength windows of integration for obtaining spectrally resolved intensity distributions centered at 629.3 nm, 630.6 nm, 632.4 nm and 633.2 nm, respectively.

found in N adjacent equally wide spectral regions, each value is then centered in its region and passed to a Gaussian fit function. Results shown here were evaluated for N = 16.

If no fitting and subtraction procedure were performed, differences between maps would be hardly distinguishable, due to the large contribution of the background with respect to the relatively small intensities variations originated by the narrow peaks. We show in Fig. S7, as an example, the results of the fitting and subtraction procedure when applied to the spectrum measured in point A.

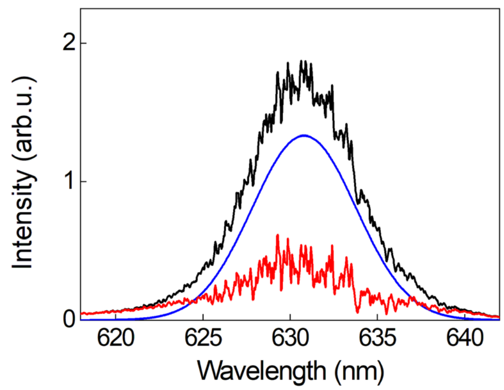

Figure S7. | Subtraction of the gaussian background. Measured spectrum collected at position A (black line), fitted Gaussian curve to local minima (blue line), spectrum obtained after background subtraction (red line).

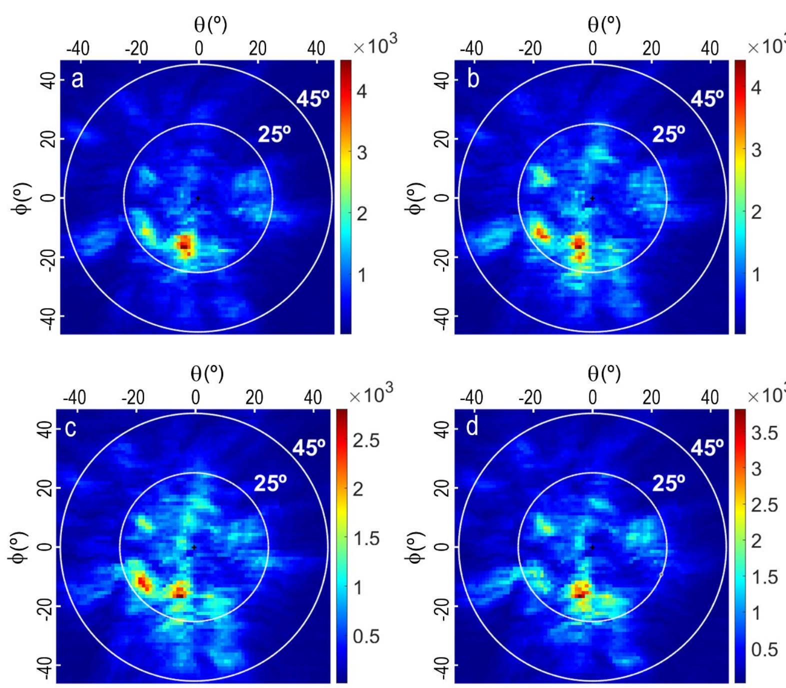

Figure S8. | Spectrally resolved far field angular intensity distributions. (a) Intensity distribution map obtained by integrating in 0.25 nm windows centered at 629.3 nm (a) 630.6 nm (b) 632.4 nm (c) and 633.2 nm (d).

In Fig. S8, we show the intensity maps obtained by integrating the collected spectra over 0.25 nm wide wavelength regions centred at 629.3 nm, 630.6 nm, 632.4 nm and 633.2 nm, after performing the background subtraction. We calculated the corresponding 2D correlation coefficients for each pair of maps, defined as $C_{m,n}$, where sub-indices *m* and *n* refers to the central wavelength at which the spectral map is resolved: 629.3 nm (1), 630.6 nm (2), 632.4 nm (3) and 633.2 nm (4), corresponding to Fig. S8 (a), (b), (c) and (d), respectively. We obtain values between 0.89 and 0.94, with $C_{1,2}$ = 0.93, $C_{1,3}$ = 0.89, $C_{1,4}$ = 0.9, $C_{2,3}$ = 0.94, $C_{2,4}$ = 0.92, $C_{3,4}$ = 0.92.

Additional measurements are performed by shining the collimated output beam of the device under study onto an opaque adhesive tape, used as a diffuser. Scattered light from the adhesive tape illuminates a resolution test target, which is imaged on an 8 bit CCD camera. In Fig. S9, we show the results obtained below and above threshold, with respect to the threshold current I_{TH}: with a bias current $I = 0.6\ I_{TH}$, for the original and the modified laser diode, in Fig. S9(a) and (b), respectively, and for a bias current $I = 1.2\ I_{TH}$, for the original and the modified laser diode, in Fig. S9(c) and (d), respectively. We observe that the modified laser diode provides a lower contrast compared to the original device, both below and above threshold as also shown with the fringe visibility and the speckle contrast reported in the main manuscript.

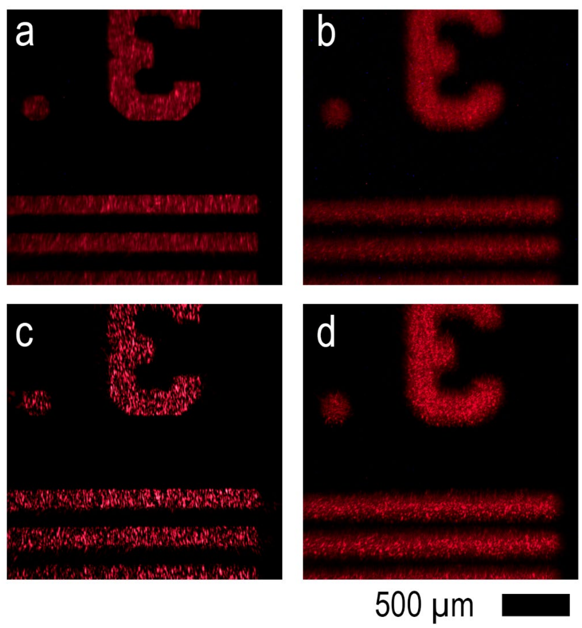

Figure S9. | Imaging of the resolution test target with original and modified laser diode. Images obtained with bias current I = 0.6 I_{TH}, with the original (a) and modified (b) laser diode, and with bias current I = 1.2 I_{TH}, with the original (c) and modified (d) laser diode.

Section 5. Theoretical model and numerical simulations

RLs with non-distributed feedback can be treated as FP lasers with scattering mirrors, with frequency dependent amplitude and phase profiles. We introduced this simple model in [1] where the scattering media were two scattering volumes, i.e. agglomerations of TiO_2 nanoparticles, enclosing a liquid dye solution. In our model, a lasing mode is allowed to exist at a given frequency if a closed loop formed between the two scattering mirrors accumulates $m\pi$ optical phase taking into account the random phase contributions of scatters at that frequency, with m an integer number. The same behavior has been observed in devices with scattering surfaces, given by the defects at the interface between air and a dye doped polymer [2,3]

In this manuscript, the modified laser diode consists of a gain medium (quantum wells) placed between the unmodified flat back mirror and the ablated rough front mirror of the original device. Accordingly, we run sets of simulations for the original and modified laser diode with spectral responses $R_{1,2}(\nu)\cdot\exp(i\phi_{1,2}(\nu))$, where ν is the frequency and $R_{1,2}(\nu)$ and $\phi_{1,2}(\nu)$ are the amplitude and phase of mirror 1 (front) and 2 (back), respectively.

In all simulations, we consider for the back flat mirror $R_2(\nu)$ and $\phi_2(\nu)$ as constant with frequency and their values are fixed to: $R_2(\nu) = 0.9$ and $\phi_2(\nu) = 0$. For the original FP laser diode, $R_1(\nu) = 0.3$ and $\phi_1(\nu) = 0$, for all frequencies (we assume a lower value of reflectivity for mirror 1 as being the output mirror). For the modified laser diode, we consider both uniform and Gaussian statistical distributions for $R_1(\nu)$ and $\phi_1(\nu)$. The results obtained with uniform distributions are presented in the main manuscript, with $R_1(\nu)$ ranging between 0.1 and 0.3 and $\phi_1(\nu)$ ranging between $-\pi$ and $+\pi$, their probability density functions are plotted in Fig. S10 (a) and (b) and their spectral profiles in Fig. S10 (c) and (d).

Additionally, we present here the results obtained when $R_1(\nu)$ and $\phi_1(\nu)$ have Gaussian distributions, with mean values μ_{R1} and $\mu_{\phi 1}$ and standard deviations σ_{R1} and $\sigma_{\phi 1}$, respectively. In Fig. S10 (e) and (f), we show the probability density functions of $R_1(\nu)$, with $\mu_{R1} = 0.2$ and $\sigma_{R1} = 0.03$, and of $\phi_1(\nu)$, with $\mu_{\phi 1} = 0$ and $\sigma_{\phi 1} = 0.1$, respectively. The corresponding spectral profiles are plotted in Fig. S10 (g) and (h). These cases correspond to the Gaussian distributions employed to calculate the spectral response and its Intensity Fourier Transform (IFT) that is reported in Fig. 4 c) (red line) of the main text.

In the computation, we construct a frequency vector ν with 54000 points, corresponding in wavelength to 80 nm range with 1.5 pm resolution centred at 630 nm. The gain is modelled with a gaussian profile centred at 630 nm and with FWHM of 16.5 nm. Losses and allowed frequencies are found for original and modified lasers, by solving the round trip equations with $R_{1,2}(\nu)\cdot\exp(i\phi_{1,2}(\nu))$:

$$g_{TH}(\nu) = \alpha + \frac{1}{2L} ln\left(\frac{1}{R_1(\nu)R_2(\nu)}\right) \qquad (1)$$

$$\frac{2n\nu L}{c} + \frac{\phi_1(\nu)+\phi_2(\nu)}{2\pi} = m \qquad (2)$$

where $g_{TH}(\nu)$ is the frequency dependent lasing threshold, α are the internal losses, L = 365 µm is the cavity length, n = 3.5 the refractive index, c the speed of light and m an integer number. Internal losses, $\alpha = 113\ cm^{-1}$, were estimated from the measured slope efficiency, η_{FP}, of the original laser diode [10], considering that $\eta_{FP} = (\alpha_{mir}/(\alpha_{mir}+\alpha))\cdot h\nu_0/q$, where $\alpha_{mir} = 0.5\cdot L^{-1}\cdot \ln[(R_1(\nu)\cdot R_2(\nu))^{-1}] = 18\ cm^{-1}$, with $R_1(\nu) = 0.3$ and $R_2(\nu) = 0.9$ for all ν, h is the Planck's constant, ν_0 the central frequency and q the electron charge.

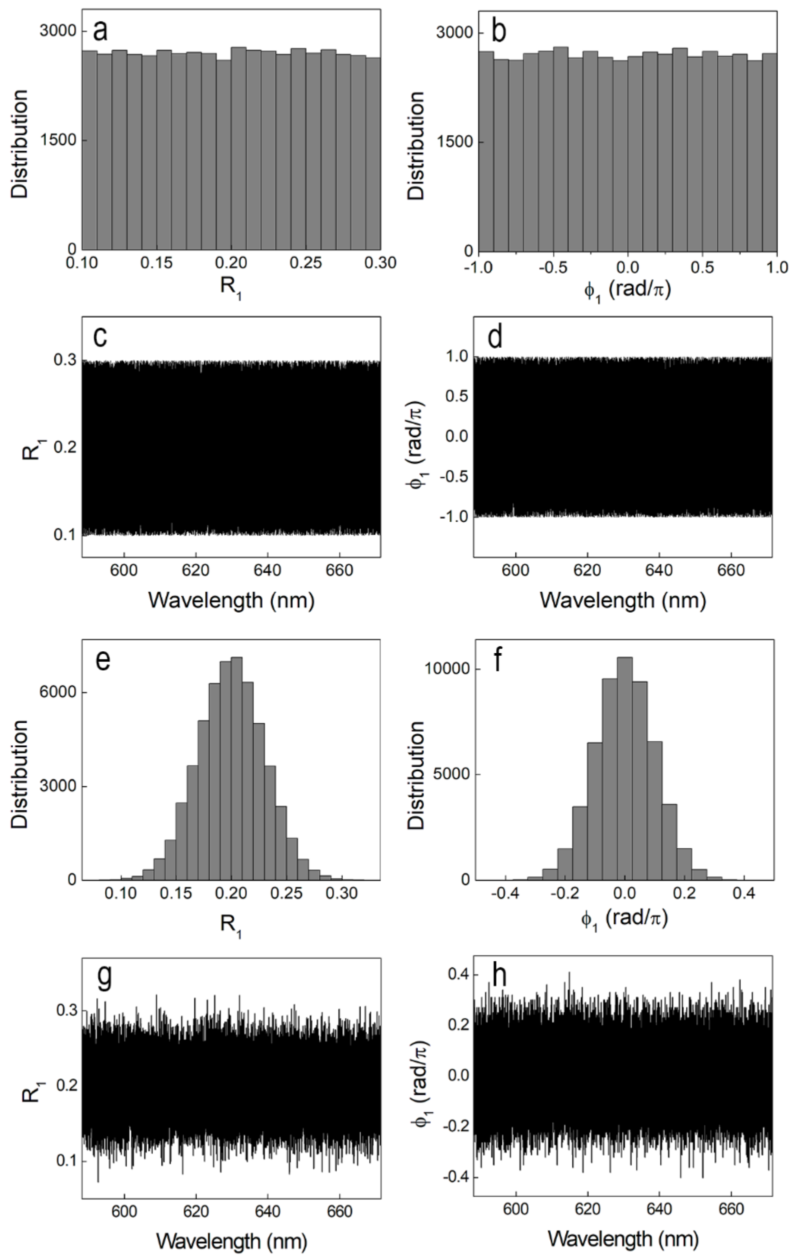

Figure S10. | Numerically constructed spectral responses of the modified front mirror. Probability density functions of amplitude (a) and phase (b) and spectral profiles of amplitude (c) and phase (d) for uniform distributions with values ranging between 0.1 and 0.3 and between -π and +π, for amplitude and phase, respectively. Probability density functions of amplitude (e) and phase (f) and spectral profiles of amplitude (g) and phase (h) for Gaussian distributions, with mean values μ_{R1} = 0.2 and $\mu_{\phi 1}$ = 0 and standard deviations σ_{R1} = 0.03 and $\sigma_{\phi 1}$ = 0.1.

Simulation results for the original laser diode are shown in Fig. S11. The simulated original laser diode shows constant losses of 131 cm^{-1}, given by the right side of Eq. (1), which corresponds to the sum of internal and constant mirror losses. By solving Eq. (2), we obtain equally spaced modes with a free spectral range of 0.155 nm, corresponding to the FP cavity length L = 365 µm.

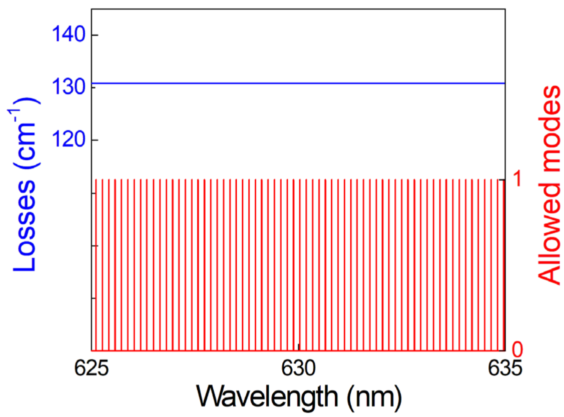

Figure S11. | Simulation results: original FP laser diode. Losses (blue line curve) and allowed modes (red line curve) of the simulated original laser diode.

Results obtained by employing uniform distributions of $R_1(\nu)$ and $\varphi_1(\nu)$ are presented in the main text, in Fig. 4 (a)-(b) and in Fig. 4 (c) (black line). Here, results obtained with Gaussian distributions of $R_1(\nu)$ and $\phi_1(\nu)$, with mean values $\mu_{R1} = 0.2$ and $\mu_{\phi 1} = 0$ and standard deviations $\sigma_{R1} = 0.03$ and $\sigma_{\phi 1} = 0.1$.are presented in Fig. S12. Losses vary with frequency, with an average value of 137 cm^{-1}, due to the contribution of $R_1(\nu)$ in α_{mir}, see the blue line curve in Fig. S12(a). Allowed modes, plotted with red line curve in Fig. S12(a), are obtained at frequencies where Eq. (2) is verified., resulting in the emission spectrum shown in Fig. S12(b).

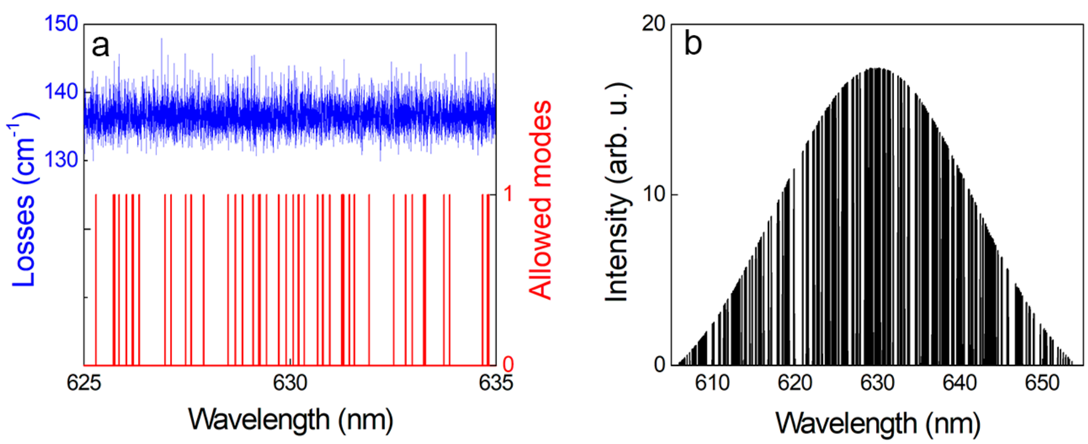

Figure S12. | Simulation results: modified laser diode with reflectivity and phase exhibiting Gaussian distribution. (a) Simulated losses (blue line, left axis) and allowed modes (red line, right axis) and (b) spectral response of the modified laser diode, with mean values $\mu_{R1} = 0.2$ and $\mu_{\phi 1} = 0$ and standard deviations $\sigma_{R1} = 0.03$ and $\sigma_{\phi 1} = 0.1$.

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